Kareem Jaber

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Education

Princeton University — Princeton, NJ
A.B. in Mathematics, GPA 3.975 — Aug. 2022 – Present
Expected minor in Computer Science
Relevant coursework:

- **Analysis**: Complex Analysis, Measure Theory, Functional Analysis, Fourier and Harmonic Analysis (independent work)
- **Algebra**: Linear and Abstract Algebra, Commutative Algebra, Representation Theory, Algebraic Number Theory
- **Topology and Geometry**: Point set Topology, Algebraic Topology, Differential Geometry (independent work), Riemann Surfaces, Algebraic Geometry (independent work)
- **Combinatorics and Probability**: The Probabilistic Method, Algebraic Combinatorics (independent work)
- **Programming**: Systems programming, Functional programming (in OCaml)

Thomas Jefferson High School for Science and Technology — Alexandria, VA
GPA 4.450, SAT 1550 — Aug. 2018 – Jun. 2022

Research and Relevant Experience

Geometric Measure Theory Research | Princeton, NJ — Summer 2025

- Worked under Camillo De Lellis on improving best known results on the Besicovitch 1/2 conjecture, in both computational and theoretical directions

Algebraic Geometry Independent Work | Princeton, NJ — Summer 2025

- Worked under Jakub Witaszek in learning the foundations of scheme theory and coherent sheaf cohomology, in preparation for my senior thesis

Junior Independent Work | Princeton, NJ — Spring 2025

- Wrote an exposition of work of Stein, Bourgain, Carbery, and Muller, on dimension independent estimates for the Hardy-Littlewood maximal function associated to a convex body. This project was advised by Assaf Naor, and my paper can be found here.

SMALL Williams REU | Willams, MA — Summer 2024

- Researched point configurations over finite fields with Professors Eyvindur Palsson and Alex Iosevich, resulting in a paper and a talk at the 2024 Young Mathematicians Conference, as well as the 2025 Joint Mathematics Meetings.
- Researched the distribution of low-lying zeros of L-functions with Prof. Steven J. Miller, resulting in a paper and a talk at the 2025 Joint Mathematics Meetings.

Undergraduate Course Assistant | Princeton Intro Analysis Sequence — Sept. 2023 – Present

- Assist with material in set theory, linear algebra, topology, metric spaces, multivariable analysis, differential geometry, and differential topology.
- Help first years form a supportive community as they enter university
- Write companion notes for the course material to aid students' understanding and intuition

Engineering Teacher | Kuching, Malaysia; International Internship Program — Summer 2023

- Worked at Chumbaka, an engineering education company in Malaysia; co-taught their coding and electronics curriculum in after-school programs; trained teachers in using material in their classrooms; contributed to company-wide curriculum development and R&D meetings
- Assisted in adapting Chumbaka's "Coding and Algorithm" course for offline use, hosting material on a Raspberry Pi to make accessible to rural schools
- Mentored a group of secondary students designing a device to track hemorrhagic shock treatment information in a hospital environment; students won first place in regional competition and are now working closely with Sarawak emergency room doctors

Combinatorial Knot Theory Research | Polymath Jr REU — Summer 2022

- Collaborated virtually for 8 weeks under the guidance of Prof. Alex Zupan, University of Nebraska–Lincoln, focusing on n-colorings of symmetric union presentations, and resulting in a paper

Mathematics and Education Outreach — Summer 2020 - present

- Established my own math circle after-school program at several local elementary schools in the Princeton community
- Attended the Math Circle Institute 2025 at Notre Dame to share my outreach experiences and learn from professional groups involved with mathematics outreach globally
- Organize and administer Princeton Splash, an annual education outreach event for high school students, and as represented Princeton at SplashCon 2024, a conference at MIT to learn more about effective education outreach
- Write expository mathematics papers on problem solving and undergraduate mathematics for high schoolers

Other Leadership Roles

Princeton University Residential College Advisor, act as the first line of support for a group of freshman as they acclimate to college

Outdoor Action Orientation Leader lead backpacking trips for incoming freshmen, certified in Wilderness Advanced First Aid

Awards and Honors

Princeton University Andrew H. Brown Prize 2025, awarded to outstanding juniors in mathematics

Putnam 2023 Top 500

Princeton University Shapiro Prize For Academic Excellence 2023 (awarded to top 3% of students in graduating class)

Runner up in Steven H. Strogatz Prize for Math Communication 2022 for paper "The Unsolvable Configuration of the 15 Puzzle and an Interesting Approach to Abstract Algebra"

Classiq Quantum Computing Competition, 1st Place Youth Prize and 3rd Place Overall, 2022

Two-time AIME qualifier, 2021 and 2022

Skills and Interests

Programming: Proficient in Python, C, Ocaml, knowledgeable in C++, Java, Haskell, web development

Personal Skills: Reflective and adaptive, creative and unconventional, curious and inquisitive

Fun Interests: Piano performer and composer, 3D printed puzzle designer, rock climber, rubber duck collector